Steven M. Wise

Professor of Mathematics

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Updated September 2026

Current Appointment

Department of Mathematics, University of Tennessee, Knoxville

Professor, August 2017 – present
Associate Head and Director of Graduate Studies, April 2025 – present
Interim Department Head, January 2020
Associate Professor, August 2012 – July 2017
Assistant Professor, August 2007 – July 2012

Research Impact

Web of Science Highly Cited Researcher: Mathematics, 2020, 2022
https://recognition.webofscience.com/awards/highly-cited/2020/
https://recognition.webofscience.com/awards/highly-cited/2022/

Google Scholar Citation Metrics: more than 11,100 citations; h-index: 55 (May 2026)

Research Interests

Computational Mathematics and Numerical Analysis: Approximation methods for nonlinear parabolic and hyperbolic PDE; convex-concave decomposition methods for bistable gradient equations; multilevel preconditioners; fast nonlinear solvers; efficient adaptive multigrid methods for interface and moving-boundary problems, especially phase-field and diffuse-interface methods; and rigorous stability and convergence analysis of finite-difference, finite element, pseudospectral, and discontinuous Galerkin methods for nonlinear, nonlocal, elliptic, and parabolic PDE.

Optimization: Preconditioned Nesterov and multilevel methods for constrained and unconstrained optimization problems, especially those arising from the discretization of nonlinear PDE and problems in the calculus of variations.

Mathematical Biology: Modeling tissue growth and remodeling, including cancerous tumor growth, angiogenesis, and cancer treatment; spatiotemporal pattern formation and morphogenesis in biological systems; and the deformation, decomposition, and motion of biological membranes in viscous flows.

Computational Materials Science: Growth and coarsening dynamics of heteroepitaxial thin films; long-time, large-scale coarsening of material microstructures; quantum-dot formation; phase-field-crystal models; nonlocal models of phase transformation; crystal growth with strong interfacial anisotropy; elastically mediated phase transformations; and soft materials.

Computational Fluid Dynamics: Phase-field methods for two-phase flows with large density differences and heat transport; two-phase flows in porous media; microstructural evolution of multiphase polymer systems, especially for organic photovoltaic devices; ion transport in solution; and multispecies gas dynamics and Boltzmann equations.

Nonlinear PDE: Existence, uniqueness, and regularity of solutions to nonlinear parabolic-hyperbolic PDE and related integro-partial differential equations, especially for models of the phenomena described above.

Nonlinear Analysis: Γ-convergence and asymptotic analysis of diffuse-interface and diffuse-domain problems.

Education

Ph.D., University of Virginia, Engineering Physics, May 2003.
Advisor: W.C. Johnson
Dissertation: “Diffuse Interface Model for Microstructural Evolution of Stressed, Binary Thin Films on Patterned Substrates."
Interdisciplinary applied-physics program; research focused on the theory and simulation of phase transformations.

Graduate Studies, Virginia Tech and Penn State, Mathematics, August 1996 – December 1998.
Coursework: real, complex, functional, and numerical analysis; applied mathematics.

B.S., Clarion University of Pennsylvania, Mathematics, May 1996.
Minor equivalent: Physics

Current Funding

NSF Grant, DMS-2607995: PI, *Diffuse Domain Methods for One and Two-Sided Elliptic Interface Problems and Applications,* $350,000, 8/26 – 7/29.

NSF Grant, DMS-2309547: PI, *Collaborative Research: Accurate and Structure-Preserving Numerical Schemes for Variable Temperature Phase Field Models and Efficient Solvers,* $213,743, 8/23 – 7/26.

ORNL-DOE-UT-Battelle CW24420: PI, *BGK Kinetic Equations,* $606,000, 8/24 – 7/27. ORNL Technical Point of Contact: Daniel A. Reasor, reasorda@ornl.gov.

Previous Funding, Support, and Awards

NSF Grant, DMS-2012634: PI, *Collaborative Research: Efficient, Accurate, and Structure-Preserving Numerical Methods for Phase Field-Type Models with Applications,* $200,000, 8/20 – 7/24.

ORNL-DOE-UT-Battelle CW24420: PI, *BGK Kinetic Equations,* $250,200, 1/22 – 7/24. ORNL Technical Point of Contact: Ryan Glasby, glasbyrs@ornl.gov.

LANL-ORNL-DOE-UT-Battelle RFP584197: Co-PI, *Developing a Better Understanding of Adaptive Velocity Grids for Kinetic Equations,* approximately $60,000, 2/20 – 5/21. DOE-LANL Technical Point of Contact: Jeff Haack. (Originally funded at approximately $640,000 before budget reductions.)

ORNL-DOE-UT-Battelle 3400267345: PI, *Quadratic Unconstrained Binary Optimization and Regularized Problems Expressed via Steady States of Dynamical Systems,* $40,506, 6/18 – 8/18. ORNL Technical Point of Contact: Ryan Bennink.

ORNL-DOE-UT-Battelle 3400253096: PI, *Physics-Based, Self-Consistent Simulation of Neutron Reflectivity for Thin Films of Block Copolymers with Incorporated Nanoparticles,* $138,834, 7/17 – 9/18. ORNL Technical Points of Contact: Rajeev Kumar, Valeria Lauter, Bobby Sumpter.

Dresden Fellowship: Technische Universität Dresden, Germany, € 10,500, 9/1/17 – 11/30/17.

ORNL-DOE-UT-Battelle 3400252731: PI, *Modeling and Simulation of Ion Transport in Polymer Systems and Novel Discrete Ordinate Methods for Modeling Radiation Transport via Kinetic Equations,* $60,285, 6/17 – 5/18. ORNL Technical Point of Contact: Cory Hauck.

IMA Conference Grant: Co-PI, *The 2017 Barrett Lectures: The Mathematical Foundations of Data Science,* $5,000, 4/17 – 3/18.

NSF Grant, DMS-1719854: PI, *Efficient, Adaptive, and Convergent Numerical Methods for Phase Field Equations with Applications,* $200,000, 8/17 – 11/21.

NSF Grant, DMS-1700494: Co-PI, *The 2017 Barrett Lectures: The Mathematical Foundations of Data Science,* $16,000, 4/17 – 3/18.

NSF Grant, DMS-1418692: PI, *Efficient, Adaptive, and Convergent Numerical Methods for Phase Field and Phase Field Crystal Equations with Applications,* $105,000, 9/14 – 8/18.

NSF Grant, DMS-1338314: Co-PI, *Southeastern-Atlantic Regional Conference on Differential Equations 2013,* $21,938, 8/13 – 4/14.

NSF Grant, DMS-1115390: Lead PI, *Collaborative Research: Stable and Efficient Convexity-Splitting Schemes for Bistable Gradient PDEs,* $160,000, 7/11 – 6/14.

NSF Grant, DMS-0818030: PI, *Collaborative Research: Multiscale Modeling of Solid Tumor Growth,* $70,000, 9/08 – 8/11.

Workshop Support: Lead Organizer, *Investigative Workshop: Solid Tumor Modeling*, National Institute for Mathematical and Biological Synthesis (NIMBioS), approximately 40 participants, 1/19/11 – 1/21/11.

Excellence in Research and Creative Achievement Award: University of Tennessee, Spring 2025, $3,000.

Senior Research and Creative Achievement Award: College of Arts and Sciences, Natural Sciences Category, University of Tennessee, Fall/Winter Convocation 2022.

Mid-Career Faculty Research Achievement Award: College of Arts and Sciences, University of Tennessee, Fall/Winter Convocation 2014, $500.

Junior Faculty Research Achievement Award: College of Arts and Sciences, University of Tennessee, Fall/Winter Convocation 2010, $1,000.

Best Graduate Teaching Award: Mathematics Department, University of Tennessee, 2012-13, and 2013-14.

Best Graduate Advising Award: Mathematics Department, University of Tennessee, 2016-17.

Supplementary Support for PhD Students

These awards are included because the PI helped prepare the applications and supervised the funded work. Awards were made directly to the students.

Charlotte Beckford, University of Tennessee-Oak Ridge Innovation Institute (UT-ORII) Science Alliance Graduate Advancement, Training and Education (GATE) Fellowship, approximately $37,000 per year plus tuition, 8/2024 – 7/2026. Topic: “Thermodynamically Consistent Models of Polymeric Battery Systems." Advisors: R. Kumar (ORNL) and **S.M. Wise** (UTK).

Kylie Berry, U.S. Department of Energy Office of Science Graduate Student Research (SCGSR) Program Award/Fellowship, approximately $18,000 plus tuition, 1/19 – 6/19. Ph.D., 2019. Topic: “Energy-Based Blending for Local-Nonlocal Coupling in Linear Elasticity." Advisors: P. Seleson (ORNL), Tadele Mengesha (UTK), and **S.M. Wise** (UTK).

John Cummings, U.S. Department of Energy Office of Science Graduate Student Research (SCGSR) Program Award/Fellowship, approximately $36,000 plus tuition, 1/15 – 1/16. Ph.D., 2017. Topic: “Simulating Polymer-Solvent Mixtures for Organic Photovoltaic Devices." Co-advisors: R. Kumar (ORNL) and **S.M. Wise** (UTK).

Sam Gruber, University of Tennessee-Oak Ridge Innovation Institute (UT-ORII) Science Alliance Graduate Advancement, Training and Education (GATE) Fellowship, approximately $37,000 per year plus tuition, 8/2025 – 7/2027. Topic: “Phase Transformations in $SrCoO_x$ Materials for Neuromorphic Computing Architectures." Advisors: V. Starchenko (ORNL), P. Ganesh (ORNL), and **S.M. Wise** (UTK).

Professional Membership

Society for Industrial and Applied Mathematics (SIAM)

Activity Group: Mathematical Aspects of Materials Science

Prior and Visiting Appointments

Dresden Senior Fellow, Technische Universität Dresden, September 2017 – November 2017.

Visiting Associate Researcher, Department of Mathematics, University of California, Irvine, March 2015 – June 2015.

Visiting Assistant Professor, Department of Mathematics, University of California, Irvine, July 2006 – July 2007.

Postdoctoral Researcher, Department of Mathematics, University of California, Irvine, July 2004 – June 2006.

Advisor: John S. Lowengrub

Postdoctoral Researcher, Departments of Biomedical Engineering and Mathematics, University of California, Irvine, March 2003 – June 2004.

Advisors: John S. Lowengrub and Vittorio Cristini

Research Assistant, University of Virginia, Engineering Physics Program, August 1999 – March 2003.

Scientist, U.S. Navy, NSWC-Dahlgren, Dahlgren, Virginia, January 1999 – August 1999.

Teaching Assistant and Primary Instructor, Department of Mathematics, Penn State, August 1998 – December 1998.

Research Assistant, Department of Mathematics, Virginia Tech, May 1997 – August 1998.

Teaching Assistant, Department of Mathematics, Virginia Tech, August 1996 – April 1997.

Textbooks and Research Monographs

1. A.J. Salgado and **S.M. Wise**, *Classical Numerical Analysis: A Comprehensive Course*, Cambridge University Press (2023), 937 pages.
   https://www.cambridge.org/9781108837705
   https://doi.org/10.1017/9781108942607
2. A.J. Salgado, **S.M. Wise**, and C. Wong, *Multigrid Methods: Axiomatic Convergence Theory for Linear and Weakly Nonlinear Problems*, De Gruyter (2026), 470 pages.
   https://doi.org/10.1515/9783111354880
3. T. Luong, T. Mengesha, and **S.M. Wise**, *Asymptotic Analysis and Perturbation Theory: Matched Asymptotic, Multiple Scale, and Homogenization Methods*, working title, Cambridge University Press, Texts in Applied Mathematics Series (under contract) approximately 600 pages.
4. **S.M. Wise**, *Non-Equilibrium Thermodynamics*, in preparation (20% complete), approximately 200 pages.

Journal Publications

1. D. Schaal and **S.M. Wise**, *Rado Numbers for Some Inequalities and an Arbitrary Number of Colors*, Congr. Numer. 121 (1996), 147-153.
2. **S.M. Wise**, A.J. Sommese, and L.T. Watson, *Algorithm 801: POLSYS_PLP: A Partitioned Linear Product Homotopy Code for Solving Polynomial Systems of Equations*, ACM TOMS 26 (2000) 176-200.
   https://doi.org/10.1145/347837.347885
3. W.C. Johnson, P.H. Leo, Y. Zhen, and **S.M. Wise**, *Spinodal Decomposition in Thin Plates Subjected to a Temperature Gradient*, Modeling the Performance of Engineering Structural Materials II (TMS Fall 2001 Meeting), (2001).
4. **S.M. Wise** and W.C. Johnson, *Competitive Phase Growth in Stressed Thin Films*, Modeling the Performance of Engineering Structural Materials II (TMS Fall 2001 Meeting), (2001).
5. W.C. Johnson and **S.M. Wise**, *Phase Decomposition of a Binary Thin Film on a Patterned Substrate*, Appl. Phys. Lettr. 81 (2002), 919-921.
   https://doi.org/10.1063/1.1497193
6. W.C. Johnson, **S.M. Wise**, J.Y. Huh, and J. Favergeon, *Effect of Interfacial Segregation on Phase Decomposition of a Thin Film on a Patterned Substrate*, Met. Mater. Int. 9 (2003) 1-8.
   https://doi.org/10.1007/BF03027222
7. **S.M. Wise** and W.C. Johnson, *Numerical Simulations of Pattern-Directed Phase Decomposition in a Stressed, Binary Thin Film*, J. Appl. Phys. 94 (2003) 889-898.
   https://doi.org/10.1063/1.1577230
8. **S.M. Wise**, J.S. Kim, and W.C. Johnson, *Surface-Directed Spinodal Decomposition in a Stressed Two-Dimensional Thin Film*, Thin Solid Films 473 (2004) 151-163.
   https://doi.org/10.1016/j.tsf.2004.07.075
9. **S.M. Wise**, J.S. Lowengrub, J.S. Kim, and W.C. Johnson, *Efficient Phase-Field Simulation of Quantum Dot Formation in a Strained Heteroepitaxial Film*, Superlattices and Microstructures 36 (2004) 293-304.
   https://doi.org/10.1016/j.spmi.2004.08.029

10. J.S. Lowengrub, Z. Hu, **S.M. Wise**, J.S. Kim, and A. Voigt, *Phase-Field Modeling of Step Dynamics*, MRS Proceedings, 859E (2005) JJ8.6.1-JJ8.6.6.
11. X. Zheng, **S.M. Wise**, and V. Cristini, *Nonlinear Simulation of Tumor Necrosis, Neo-Vascularization and Tissue Invasion via an Adaptive Finite-Element/Level-Set Method*, Bull. Math. Biol. 67 (2005) 211-259. https://doi.org/10.1016/j.bulm.2004.08.001
12. **S.M. Wise**, J.S. Lowengrub, J.S. Kim, K. Thornton, P. Voorhees, and W.C. Johnson, *Quantum Dot Formation on a Strain-Patterned Epitaxial Thin Film*, Appl. Phys. Lettr. 87 (2005) 133102. https://doi.org/10.1063/1.2061852
13. J. Favergeon, J.Y. Huh, W.C. Johnson, and **S.M. Wise**, *Wetting Transitions in a Binary Thin Film*, Met. Mater. Int. 11 (2005) 487-497. https://doi.org/10.1007/BF03027499
14. H.B. Frieboes, J.S. Lowengrub, **S.M. Wise**, X. Zheng, P. Macklin, E.L. Bearer, and V. Cristini, *Computer Simulation of Glioma Growth and Morphology*, NeuroImage 37 (2007) S59-S70. https://doi.org/10.1016/j.neuroimage.2007.03.008
15. **S.M. Wise**, J.S. Kim, and J.S. Lowengrub, *Solving the Regularized, Strongly Anisotropic Cahn-Hilliard Equation by an Adaptive Nonlinear Multigrid Method*, J. Comput. Phys. 226 (2007) 414-446. https://doi.org/10.1016/j.jcp.2007.04.020
16. S. Torabi, **S.M. Wise**, J.S. Lowengrub, A. Ratz, and A. Voigt, *A New Method for Simulating Strongly Anisotropic Cahn-Hilliard Equations*, MS&T 2007 Conference Proceedings (2007) 1432-1444.
17. Z. Hu, S. Li, J.S. Lowengrub, **S.M. Wise**, and A. Voigt, *Phase Field Modeling of Nanoscale Island Dynamics*, TMS 2008 (137th annual meeting) Supplemental Proceedings: Materials Processing and Properties (2008) 111-116.
18. V. Cristini, H.B. Frieboes, X. Li, J.S. Lowengrub, P. Macklin, S. Sanga, **S.M. Wise**, and X. Zheng, *Nonlinear Modeling and Simulation of Tumor Growth, in N. Bellomo, M. Chaplain, and E. De Angelis (Editors), "Selected Topics in Cancer Modeling: Genesis, Evolution, Immune Competition, and Therapy,"* Birkhauser (2008) 1-69. https://doi.org/10.1007/978-0-8176-4713-1_6
19. S. Torabi, **S.M. Wise**, S. Li, A. Voigt, J.S. Lowengrub, and P. Zhou, *Simulations of Nonlinear Strongly Anisotropic, Misfitting Crystals and Thin Films*, MRS Proceedings 1087 (2008) 1087-V02-01. https://doi.org/10.1557/PROC-1087-V02-01
20. **S.M. Wise**, J.S. Lowengrub, H.B. Frieboes, and V. Cristini, *Three-Dimensional Multispecies Nonlinear Tumor Growth–I: Model and Numerical Method*, J. Theor. Biol. 253 (2008) 524-543. https://doi.org/10.1016/j.jtbi.2008.03.027
21. V. Cristini, X. Li, J.S. Lowengrub, and **S.M. Wise**, *Nonlinear Simulations of Solid Tumor Growth using a Mixture Model: Invasion and Branching*, J. Math. Biol. 58 (2009) 723-763. https://doi.org/10.1007/s00285-008-0215-x
22. S. Torabi, J.S. Lowengrub, A. Voigt, and **S.M. Wise**, *A New Phase-Field Model for Strongly Anisotropic Systems*, Proc. R. Soc. A 465 (2009) 1337-1359. https://doi.org/10.1098/rspa.2008.0385
23. E.L. Bearer, J.S. Lowengrub, H.B. Frieboes, Y.L. Chuang, F. Jin, **S.M. Wise**, M. Ferrari, D.B. Agus, and V. Cristini, *Multiparameter Computational Modeling of Tumor Invasion*, Cancer Res. 69 (2009) 4493-4501. https://doi.org/10.1158/0008-5472.CAN-08-3834
24. J.S. Lowengrub, V. Cristini, H.B. Frieboes, X. Li, P. Macklin, S. Sanga, **S.M. Wise**, and X. Zheng, *Lecture Notes on Nonlinear Tumor Growth: Modeling and Simulation, in B.C. Khoo, Z. Li, and P. Lin (Editors) "Interface Problems and Methods in Biological and Physical Flows,"* World Scientific (2009) 69-133. https://doi.org/10.1142/9789812837851_0002
25. **S.M. Wise**, C. Wang, and J.S. Lowengrub, *An Energy Stable and Convergent Finite Difference Scheme for the Phase Field Crystal Equation*, SINUM (SIAM J. Numer. Anal.) 47 (2009) 2269-2288. https://doi.org/10.1137/080738143
26. Z. Hu, **S.M. Wise**, C. Wang, and J.S. Lowengrub, *Stable Finite Difference, Nonlinear Multigrid Simulation of the Phase Field Crystal Equation*, J. Comput. Phys. 228 (2009) 5323-5339. https://doi.org/10.1016/j.jcp.2009.04.020

27. P. Zhou, J.S. Lowengrub, and **S.M. Wise**, *Coarsening of 3D Thin Films Under the Influence of Strong Surface Anisotropy and Elastic Stresses*, TMS 2009 (138th annual meeting), Supplemental Proceedings: Materials Characterization, Computation and Modeling (2009) 39-46.
https://apps.dtic.mil/dtic/tr/fulltext/u2/a538820.pdf

28. J.S. Lowengrub, H.B. Frieboes, F. Jin, Y.L. Chuang, X. Li, P. Macklin, **S.M. Wise**, and V. Cristini, *Nonlinear Modeling of Cancer: Bridging the Gap Between Cells and Tumors*, Nonlinearity 23 (2010) R1-R91.
https://doi.org/10.1088/0951-7715/23/1/R01

29. C. Wang, X. Wang, and **S.M. Wise**, *Unconditionally Stable Schemes for Equations of Thin Film Epitaxy*, Discrete Contin. Dyn. Syst. Ser. A 28 (2010) 405-423.
https://doi.org/10.3934/dcds.2010.28.405

30. **S.M. Wise**, *Unconditionally Stable Finite Difference, Nonlinear Multigrid Simulation of the Cahn-Hilliard-Hele-Shaw System of Equations*, J. Sci. Comput. 44 (2010) 38-68.
https://doi.org/10.1007/s10915-010-9363-4

31. H.B. Frieboes, F. Jin, Y.L. Chuang, **S.M. Wise**, J.S. Lowengrub, and V. Cristini, *Three-Dimensional Multispecies Nonlinear Tumor Growth–II: Angiogenesis and Tissue Invasion*, J. Theor. Biol. 264 (2010) 1254-1278.
https://doi.org/10.1016/j.jtbi.2010.02.036

32. Y.L. Chuang, F. Jin, and **S.M. Wise**, *Numerical Schemes, in V. Cristini and J.S. Lowengrub (Authors), "Multiscale Modeling of Cancer: An Integrated Experimental and Mathematical Modeling Approach,"* Cambridge University Press (2010) 153-182.
https://doi.org/10.1017/CBO9780511781452.009

33. C. Wang and **S.M. Wise**, *Global Smooth Solutions of the Three-Dimensional Modified Phase Field Crystal Equation*, Methods Appl. Anal. 17 (2010) 191-212.
https://doi.org/10.4310/MAA.2010.v17.n2.a4

34. **S.M. Wise**, J.S. Lowengrub, and V. Cristini, *An Adaptive Algorithm for Simulating Solid Tumor Growth using Mixture Models*, Math. Comput. Model. 53 (2011) 1-20.
https://doi.org/10.1016/j.mcm.2010.07.007

35. C. Wang and **S.M. Wise**, *An Energy Stable and Convergent Finite Difference Scheme for the Modified Phase Field Crystal Equation*, SINUM (SIAM J. Numer. Anal.) 49 (2011) 945-969.
https://doi.org/10.1137/090752675

36. Z. Hu, J.S. Lowengrub, **S.M. Wise**, and A. Voigt, *Phase-Field Modeling of Epitaxial Growth: Applications to Step Trains and Island Dynamics*, Physica D 241 (2012) 77-94.
https://doi.org/10.1016/j.physd.2011.09.004

37. J. Shen, C. Wang, X. Wang, and **S.M. Wise**, *Second-Order Convex Splitting Schemes for Gradient Flows with Ehrlich-Schwoebel Type Energy: Application to Thin Film Epitaxy*, SINUM (SIAM J. Numer. Anal.) 50 (2012) 105-125.
https://doi.org/10.1137/110822839

38. X. Feng and **S.M. Wise**, *Analysis of a Darcy-Cahn-Hilliard Diffuse Interface Model for the Hele-Shaw Flow and its Fully Discrete Finite Element Approximation*, SINUM (SIAM J. Numer. Anal.) 50 (2012) 1320-1343.
https://doi.org/10.1137/110827119

39. P. Zhou, **S.M. Wise**, X. Li, and J.S. Lowengrub, *Coarsening of Elastically Stressed, Strongly Anisotropic Driven Thin Films*, Phys. Rev. E 85 (2012) 061605.
https://doi.org/10.1103/PhysRevE.85.061605

40. W. Chen, S. Conde, C. Wang, X. Wang, and **S.M. Wise**, *A Linear Energy Stable Scheme for a Thin Film Model without Slope Selection*, J. Sci. Comput. 52 (2012) 546-562.
https://doi.org/10.1007/s10915-011-9559-2

41. C. Collins, J. Shen, and **S.M. Wise**, *An Efficient, Energy Stable Scheme for the Cahn-Hilliard-Brinkman System*, Commun. Comput. Phys. 13 (2013) 929-957.
https://doi.org/10.4208/cicp.171211.130412a

42. A. Baskaran, Z. Hu, J.S. Lowengrub, C. Wang, **S.M. Wise**, and P. Zhou, *Energy Stable and Efficient Finite-Difference Nonlinear-Multigrid Schemes for the Modified Phase Field Crystal Equation*, J. Comput. Phys. 250 (2013) 270-292.
https://doi.org/10.1016/j.jcp.2013.04.024

43. A. Aristotelous, O. Karakashian, **S.M. Wise**, *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for a Modified Cahn-Hilliard Equation and an Efficient Nonlinear Multigrid Solver*, Discrete Cont. Dyn. Syst. Series B 18 (2013) 2211-2238. https://doi.org/10.3934/dcdsb.2013.18.2211

44. A. Baskaran, J.S. Lowengrub, C. Wang, and **S.M. Wise**, *Convergence of a Second Order Convex Splitting Scheme for the Modified Phase Field Crystal Equation*, SINUM (SIAM J. Numer. Anal.) 51 (2013) 2851-2873. https://doi.org/10.1137/120880677

45. W. Chen, C. Wang, X. Wang, and **S.M. Wise**, *A Linear Iteration Algorithm for an Energy Stable Second Order Scheme for a Thin Film Model Without Slope Selection*, J. Sci. Comput. 59 (2014) 574-601. https://doi.org/10.1007/s10915-013-9774-0

46. Y. Chen, **S.M. Wise**, V.B. Shenoy, and J.S. Lowengrub, *A Stable Scheme for a Nonlinear, Multispecies Tumor Growth Model with an Elastic Membrane*, Int. J. Numer. Meth. Biomed. Engng. 30 (2014) 726-754. https://doi.org/10.1002/cnm.2624

47. Z. Guan, J.S. Lowengrub, C. Wang, and **S.M. Wise**, *Second-Order Convex Splitting Schemes for Non-local Cahn-Hilliard and Allen-Cahn Equations*, J. Comput. Phys. 277 (2014) 48-71. https://doi.org/10.1016/j.jcp.2014.08.001

48. Z. Guan, C. Wang, and **S.M. Wise**, *A Convergent Convex Splitting Scheme for the Periodic Nonlocal Cahn-Hilliard Equation*, Numer. Math. 128 (2014) 377-406. https://doi.org/10.1007/s00211-014-0608-2

49. A. Diegel, X. Feng, and **S.M. Wise**, *Analysis of a Mixed Finite Element Method for a Cahn-Hilliard-Darcy-Stokes System*, SINUM (SIAM J. Numer. Anal.) 53 (2015) 127-152. https://doi.org/10.1137/130950628

50. A. Aristotelous, O. Karakashian, and **S.M. Wise**, *Second-Order in Time, Primitive-Variable Discontinuous Galerkin Schemes for a Cahn-Hilliard Equation with a Mass Source Term*, IMA J. Numer. Anal. 35 (2015) 1167-1198. https://doi.org/10.1093/imanum/dru035

51. W. Feng, T. Lewis, and **S.M. Wise**, *Discontinuous Galerkin Derivative Operators with Applications to Second Order Elliptic Problems and Stability*, Math. Methods Appl. Sci. 38 (2015) 5160-5182. https://doi.org/10.1002/mma.3440

52. J. Guo, C. Wang, **S.M. Wise**, and X. Yue, *An H^2 Convergence of a Second-Order Convex-Splitting, Finite Difference Scheme for the Three-Dimensional Cahn-Hilliard Equation*, Commun. Math. Sci. 14 (2016) 489-515. https://doi.org/10.4310/CMS.2016.v14.n2.a8

53. Tri Sri Noor Asih, Suzanne Lenhart, **S.M. Wise**, Lina Aryati, F. Adi-Kusumo, Mardiah S. Hardianti, and Jonathan Forde, *The Dynamics of HPV Infection and Cervical Cancer Cells*, Bull. Math. Biol. 78 (2016) 4-20. https://doi.org/10.1007/s11538-015-0124-2

54. W. Chen, Y. Liu, C. Wang, and **S.M. Wise**, *An Optimal-Rate Convergence Analysis of a Fully Discrete Finite Difference Scheme for the Cahn-Hilliard-Hele-Shaw Equation*, Math. Comp. 85 (2016) 2231-2257. https://doi.org/10.1090/mcom3052

55. N. Chen, C. Wang, and **S.M. Wise**, *Global-in-time Gevrey Regularity Solution for a Class of Bistable Gradient Flows*, Discrete Cont. Dyn. Syst. Series B 21 (2016) 1689-1711. https://doi.org/10.3934/dcdsb.2016018

56. Z. Guan, V. Heinonen, J.S. Lowengrub, C. Wang, and **S.M. Wise**, *An Energy Stable, Hexagonal Finite Difference Scheme for the 2D Phase Field Crystal Amplitude Equations*, J. Comput. Phys. 321 (2016) 1026-1054. https://doi.org/10.1016/j.jcp.2016.06.007

57. A. Diegel, C. Wang, and **S.M. Wise**, *Stability and Convergence of a Second Order Mixed Finite Element Method for the Cahn-Hilliard Equation*, IMA J. Numer. Anal. 36 (2016) 1867-1897. https://doi.org/10.1093/imanum/drv065

58. K. Cheng, C. Wang, **S.M. Wise**, and Y. Yue, *A Second-Order, Weakly Energy-Stable Pseudo-Spectral Scheme for the Cahn-Hilliard Equation and its Solution by the Homogeneous Linear Iteration Method*, J. Sci. Comput. 69 (2016) 1083-1114. https://doi.org/10.1007/s10915-016-0228-3

59. W. Feng, A.J. Salgado, C. Wang, **S.M. Wise**, *Preconditioned Steepest Descent Methods for some Nonlinear Elliptic Equations Involving p-Laplacian Terms*, J. Comput. Phys. 334 (2017) 45-67. https://doi.org/10.1016/j.jcp.2016.12.046

60. Y. Liu, W. Chen, C. Wang, and **S.M. Wise**, *Error Analysis of a Mixed Finite Element Method for a Cahn-Hilliard-Hele-Shaw System*, Numer. Math. 135 (2017) 679-709.
https://doi.org/10.1007/s00211-016-0813-2

61. Z. Qiao, C. Wang, **S.M. Wise**, and Z. Zhang, *Error Analysis of an Energy Stable Finite Difference Scheme for the Epitaxial Thin Film Model with Slope Selection with an Improved Convergence Constant*, Int. J. Numer. Anal. Model. 14 (2017) 283-305.

62. Z. Guo, P. Lin, J.S. Lowengrub, and **S.M. Wise**, *Mass Conservative and Energy Stable Finite Difference Methods for the Quasi-Incompressible Navier-Stokes-Cahn-Hilliard System: Primitive Variable and Projection-Type Schemes*, Comput. Methods Appl. Mech. Engrg. 326 (2017) 144-174.
https://doi.org/10.1016/j.cma.2017.08.011

63. A. Diegel, C. Wang, X. Wang, and **S.M. Wise**, *Error Analysis of a Second Order Mixed Finite Element Method for the Cahn-Hilliard-Navier-Stokes Equation*, Numer. Math. 137 (2017) 495-534.
https://doi.org/10.1007/s00211-017-0887-5

64. W. Feng, Z. Guo, J.S. Lowengrub, and **S.M. Wise**, *A Mass-Conservative Adaptive FAS Multigrid Solver for Cell-Centered Finite Difference Methods on Block-Structured, Locally-Cartesian Grids*, J. Comput. Phys. 352 (2018) 463-497.
https://doi.org/10.1016/j.jcp.2017.09.065

65. Y. Yan, W. Chen, C. Wang, and **S.M. Wise**, *A Second-Order Energy Stable BDF Numerical Scheme for the Cahn-Hilliard Equation*, Commun. Comput. Phys. 23 (2018) 572-602.
https://doi.org/10.4208/cicp.OA-2016-0197

66. L. Dong, W. Feng, C. Wang, **S.M. Wise**, and Z. Zhang, *Convergence Analysis and Numerical Implementation of a Second Order Numerical Scheme for the Three-Dimensional Phase Field Crystal Equation*, Comput. Math. Appl. 75 (2018) 1912-1928.
https://doi.org/10.1016/j.camwa.2017.07.012

67. J. Cummings, J.S. Lowengrub, B.G. Sumpter, **S.M. Wise** and R. Kumar, *Modeling Solvent Evaporation During Thin Film Formation in Phase Separating Polymer Mixtures*, Soft Matter 14 (2018) 1833-1846.
https://doi.org/10.1039/c7sm02560b

68. L.G. Rebholz, **S.M. Wise**, and M. Xiao, *Penalty-Projection Schemes for the Cahn-Hilliard Navier-Stokes Diffuse Interface Model of Two Phase Flow, and their Connection to Divergence-Free Coupled Schemes*, Int. J. Numer. Anal. Model. 15 (2018) 649-676.

69. Y. Chen, J.S. Lowengrub, J. Shen, C. Wang, and **S.M. Wise**, *Efficient Energy Stable Schemes for Isotropic and Strongly Anisotropic Cahn-Hilliard Systems with the Willmore Regularization*, J. Comput. Phys. 365 (2018) 56-73.
https://doi.org/10.1016/j.jcp.2018.03.024

70. W. Feng, Z. Guan, J.S. Lowengrub, C. Wang, **S.M. Wise**, and Y. Chen, *An Energy Stable Finite-Difference Scheme for the Functionalized Cahn-Hilliard Equation and its Convergence Analysis*, J. Sci. Comput. 76 (2018) 1938-1967.
https://doi.org/10.1007/s10915-018-0690-1

71. W. Feng, C. Wang, **S.M. Wise**, Z. Zhang, *A Second-Order Energy Stable Backward Differentiation Formula Method for the Epitaxial Thin Film Equation with Slope Selection*, Numer. Methods Partial Differ. Eq. 34 (2018) 1975-2007.
https://doi.org/10.1002/num.22271

72. R. Backofen, **S.M. Wise**, M. Salvalaglio, and A. Voigt, *Convexity Splitting in a Phase Field Model for Surface Diffusion*, Int. J. Numer. Anal. Model. 16 (2019) 192-209.

73. W. Chen, W. Feng, Y. Liu, C. Wang, and **S.M. Wise**, *A Second Order Energy Stable Scheme for the Cahn-Hilliard-Hele-Shaw Equations*, Discrete Cont. Dyn. Syst. Series B 24 (2019) 149-182.
https://doi.org/10.3934/dcdsb.2018090

74. K. Cheng, W. Feng, C. Wang, and **S.M. Wise**, *An Energy Stable Fourth Order Finite Difference Scheme for the Cahn-Hilliard Equation*, J. Comput. Appl. Math. 362 (2019) 574-595.
https://doi.org/10.1016/j.cam.2018.05.039

75. W. Chen, C. Wang, X. Wang, and **S.M. Wise**, *Positivity-Preserving, Energy Stable Numerical Schemes for the Cahn-Hilliard Equation with Logarithmic Potential*, J. Comput. Phys. X 3 (2019) 100031 (29 pages).
https://doi.org/10.1016/j.jcpx.2019.100031

76. J.M. Church, Z. Guo, P.K. Jimack, A. Madzvamuse, K. Promislow, B. Wetton, **S.M. Wise**, and F. Yang, High Accuracy Benchmark Problems for Allen-Cahn and Cahn-Hilliard Dynamics, Commun. Comput. Phys. 26 (2019), pp. 947-972.
https://doi.org/10.4208/cicp.OA-2019-0006

77. K. Cheng, C. Wang, and **S.M. Wise**, *An Energy Stable BDF2 Fourier Pseudo-Spectral Numerical Scheme for the Square Phase Field Crystal Equation*, Commun. Comput. Phys. 26 (2019) 1335-1364.
https://doi.org/10.4208/cicp.2019.js60.10

78. K. Cheng, C. Wang, **S.M. Wise**, and Z. Yuan, *Global-in-Time Gevrey Regularity Solutions for the Functionalized Cahn-Hilliard Equation*, Discrete Cont. Dyn. Syst. Series S 13 (2020) 2211-2229.
https://doi.org/10.3934/dcdss.2020186

79. K. Cheng, C. Wang, and **S.M. Wise**, *A Weakly Nonlinear, Energy Stable Scheme for the Strongly Anisotropic Cahn-Hilliard Equation and Its Convergence Analysis*, J. Comput. Phys. 405 (2020) 109109 (28 pages).
https://doi.org/10.1016/j.jcp.2019.109109

80. L. Chen, X. Hu, and **S.M. Wise**, *Convergence Analysis of the Fast Subspace Descent Methods for Convex Optimization Problems*, Math. Comp. 89, (2020) 2249-2282.
https://doi.org/10.1090/mcom/3526

81. W. Chen, C. Wang, S. Wang, X. Wang, and **S.M. Wise**, *Energy Stable Numerical Schemes for a Ternary Cahn-Hilliard System*, J. Sci. Comp. 84 (2020) 27 (36 pages).
https://doi.org/10.1007/s10915-020-01276-z

82. C. Zhang, J. Ouyang, C. Wang, and **S.M. Wise**, *Numerical Comparison of Modified-Energy Stable SAV-Type Schemes and Classical BDF Methods on Benchmark Problems for the Functionalized Cahn-Hilliard Equation*, J. Comput. Phys. 423 (2020) 109772 (35 pages).
https://doi.org/10.1016/j.jcp.2020.109772

83. Z. Guo, F. Yu, P. Lin, **S.M. Wise**, and J.S. Lowengrub, *A Diffuse Domain Method for Two Phase Flows with Large Density Ratio in Complex Geometries*, J. Fluid Mech. 907 (2021) A38.
https://doi.org/10.1017/jfm.2020.790

84. J. Guo, C. Wang, **S.M. Wise** and X. Yue, *An Improved Error Analysis for a Second-Order Numerical Scheme for the Cahn-Hilliard Equation*, J. Comput. Appl. Math. 388 (2021). 113300.
https://doi.org/10.1016/j.cam.2020.113300

85. M. Salvalaglio, A. Voigt, and **S.M. Wise**, *Doubly Degenerate Diffuse Interface Models of Surface Diffusion*, Math. Methods Appl. Sci. 44 (2021) 5385–5405.
https://doi.org/10.1002/mma.7116

86. M. Salvalaglio, M. Selch, A. Voigt, and **S.M. Wise**, *Doubly Degenerate Diffuse Interface Models of Anisotropic Surface Diffusion*, Math. Methods Appl. Sci. 44 (2021) 5406–5417.
https://doi.org/10.1002/mma.7118

87. M. Yuan, W. Chen, C. Wang, **S.M. Wise**, and Z. Zhang, *An Energy Stable Finite Element Scheme for the Three-Component Cahn-Hilliard-Type Model for Macromolecular Microsphere Composite Hydrogels*, J. Sci. Comp. 87 (2021) 78.
https://doi.org/10.1007/s10915-021-01508-w

88. J. Zhang, C. Wang, **S.M. Wise**, and Z. Zhang, *Structure-preserving, Energy Stable Numerical Schemes for a Liquid Thin Film Coarsening Model*, SIAM J. Sci. Comput. 42 (2021) A1248–A1272.
https://doi.org/10.1137/20M1375656

89. L. Dong, C. Wang, **S.M. Wise**, and Z. Zhang, *A Positivity-Preserving, Energy Stable Scheme for a Ternary Cahn-Hilliard System with Singular Interfacial Parameters*, J. Comput. Phys. 442 (2021) 110451 (29 pages).
https://doi.org/10.1016/j.jcp.2021.110451

90. C. Liu, C. Wang, **S.M. Wise**, X. Yue, and S. Zhou, *A Positivity-Preserving, Energy Stable and Convergent Numerical Scheme for the Poisson-Nernst-Planck System*, Math. Comp. 90 (2021) 2071–2106.
https://doi.org/10.1090/mcom/3642

91. S. Sahyoun, D. Wilson, S.M. Djouadi, **S.M. Wise**, H.A. Abderrahmane, *Proper Orthogonal Decomposition Reduced Order Model for Tear Film Flows*, American Control Conference (ACC) (2021) 2763–2768.
https://doi.org/10.23919/ACC50511.2021.9483018

92. J. Park, A.J. Salgado, and **S.M. Wise**, *Preconditioned Accelerated Gradient Descent Methods for Locally Lipschitz Smooth Objectives with Applications to the Solution of Nonlinear PDEs*, J. Sci. Comput. 89 (2021) 17 (37 pages).
https://doi.org/10.1007/s10915-021-01615-8

93. W. Chen, J. Jing, C. Wang, X. Wang and **S.M. Wise**, *A Modified Crank-Nicolson Numerical Scheme for the Flory-Huggins Cahn-Hilliard Model*, Commun. Comput. Phys. 31 (2022), 60–93.
https://doi.org/10.4208/cicp.OA-2021-0074

94. C. Liu, C. Wang, **S.M. Wise**, X. Yue, and S. Zhou, *An Iterative Solver for the Poisson-Nernst-Planck System and its Convergence Analysis*, J. Comput. Appl. Math. 406 (2022) 114017 (13 pages).
https://doi.org/10.1016/j.cam.2021.114017

95. C. Liu, C. Wang, Y. Wang, and **S.M. Wise**, *Convergence Analysis of the Variational Operator Splitting Scheme for a Reaction-Diffusion System with Detailed Balance*, SINUM (SIAM J. Numer. Anal.) 60 (2022) 781–803.
https://doi.org/10.1137/21M1421283

96. K. Cheng, C. Wang, **S.M. Wise**, and Y. Wu, *A Third Order Accurate in Time, BDF-Type Energy Stable Scheme for the Cahn-Hilliard Equation*, Numer. Math. Theor. Meth. Appl. 15 (2022) 279–303.
https://doi.org/10.4208/nmtma.OA-2021-0165

97. L. Dong, C. Wang, **S.M. Wise**, and Z. Zhang, *Optimal Rate Convergence Analysis of a Numerical Scheme for the Ternary Cahn-Hilliard System with a Flory-Huggins-deGennes Energy Potential*, J. Comp. Appl. Math. 415 (2022) 114474 (18 pages).
https://doi.org/10.1016/j.cam.2022.114474

98. E. Habbershaw and **S.M. Wise**, *A Progress Report on Numerical Methods for BGK-Type Kinetic Equations*, TRACE: Faculty Publications and Other Works, Mathematics, The University of Tennessee (2022) Report number 10.
https://trace.tennessee.edu/utk_mathpubs/10/

99. M. Yuan, W. Chen, C. Wang, **S.M. Wise**, and Z. Zhang, *A Second Order Accurate in Time, Energy Stable Finite Element Scheme for the Flory-Huggins-Cahn-Hilliard Equation*, Adv. Appl. Math. Mech. 14 (2022) 1477–1508.
https://doi.org/10.4208/aamm.OA-2021-0331

100. M. Punke, **S.M. Wise**, A. Voigt, and M. Salvalaglio, *Explicit Temperature Coupling in Phase-Field Crystal Models of Solidification*, Modelling Simul. Mater. Sci. Eng. 30 (2022) 074004 (18 pages).
https://doi.org/10.1088/1361-651X/ac8abd

101. C. Wang and **S.M. Wise**, *A Thermodynamically-Consistent Phase Field Crystal Model of Solidification with Heat Flux*, J. Math. Study 55 (2022) 1–21.
https://doi.org/10.4208/jms.v55n4.22.01

102. X. Chen, C. Wang, and **S.M. Wise**, *A Preconditioned Steepest Descent Solver for the Cahn-Hilliard Equation with Variable Mobility*, Int. J. Numer. Anal. Model. 19 (2022) 839–863.
http://www.math.ualberta.ca/ijnam/Volume-19-2022/No-6-22/2022-06-06.pdf

103. J.H. Park, A.J. Salgado, and **S.M. Wise**, *Benchmark Computations of the Phase Field Crystal and Functionalized Cahn-Hilliard Equations via Fully Implicit, Nesterov Accelerated Schemes*, Commun. Comput. Phys. 33 (2023) 367–398.
https://doi.org/10.4208/cicp.OA-2022-0117

104. K. Cheng, C. Wang, and **S.M. Wise**, *An Energy Stable Finite Difference Scheme for the Ericksen-Leslie System with Penalty Function and its Optimal Rate Convergence Analysis*, Commun. Math. Sci. 21 (2023) 1135–1169.
https://doi.org/10.4310/CMS.2023.v21.n4.a10

105. M. Punke, **S.M. Wise**, A. Voigt, and M. Salvalaglio, *Improved Time Integration for Phase-Field Crystal Models of Solidification*, Proc. Appl. Math. Mech. 23 (2023) e202200112 (6 pages).
https://doi.org/10.1002/pamm.202200112

106. K. Cheng, C. Wang, and **S.M. Wise**, *High Order Accurate and Convergent Numerical Scheme for the Strongly Anisotropic Cahn-Hilliard Model*, Numer. Methods Partial Differ. Eq. 39 (2023) 4007–4029.
https://doi.org/10.1002/num.23034

107. X. Tang, S. Li, J.S. Lowengrub, and **S.M. Wise**, *Phase Field Modeling and Computation of Vesicle Growth or Shrinkage*, J. Math. Biol. 86 (2023) 97 (31 pages).
https://doi.org/10.1007/s00285-023-01928-2

108. C. Liu, C. Wang, **S.M. Wise**, X. Yue, and S. Zhou, *A Second Order Accurate, Positivity Preserving Numerical Method for the Poisson-Nernst-Planck System and Its Convergence Analysis*, J. Sci. Comput. 97 (2023) 23 (35 pages).
https://doi.org/10.1007/s10915-023-02345-9

109. S. Dai, J. Renzi, and **S.M. Wise**, *Gamma Convergence of the DeGennes-Cahn-Hilliard Energy*, Comm. Math. Sci. 21 (2023) 2131–2144.
https://doi.org/10.4310/CMS.2023.v21.n8.a3

110. Y. Guo, C. Wang, **S.M. Wise**, Z. Zhang, *Convergence Analysis of a Positivity-Preserving Numerical Scheme for the Cahn-Hilliard-Stokes System with Flory-Huggins Energy Potential*, Math. Comp. 93 (2024) 2185–2214.
https://doi.org/10.1090/mcom/3916

111. C. Wang, J. Wang, **S.M. Wise**, Z. Xia, and L. Xu, *Convergence Analysis of a Temporally Second-Order Accurate Finite Element scheme for the Cahn-Hilliard-Magnetohydrodynamics System of Equations*, J. Comput. Appl. Math. 436 (2024) 115409 (17 pages).
https://doi.org/10.1016/j.cam.2023.115409

112. E. Habbershaw and **S.M. Wise**, *Year-2 Progress Report on Numerical Methods for BGK-Type Kinetic Equations*, TRACE: Faculty Publications and Other Works, Mathematics, The University of Tennessee (2024) Report number 11.
https://trace.tennessee.edu/utk_mathpubs/11/

113. Y. Sun, J. Wu, M. Jiang, **S.M. Wise**, and Z. Guo, *A Thermodynamically Consistent Phase-Field Model and an Entropy Stable Numerical Method for Simulating Two-Phase Flows with Thermocapillary Effects*, Appl. Numer. Math. 206 (2024) 161–189.
https://doi.org/10.1016/j.apnum.2024.08.010

114. E. Habbershaw and **S.M. Wise**, *Year-4 Progress Report on Numerical Methods for BGK-Type Kinetic Equations*, TRACE: Faculty Publications and Other Works, Mathematics, The University of Tennessee (2025) Report number 14 (206 pages).
https://trace.tennessee.edu/utk_mathpubs/14/

115. M. Punke, M. Salvalaglio, A. Voigt, and **S.M. Wise**, *A Non-Isothermal Phase-Field Crystal Model with Lattice Expansion: Analysis and Benchmarks*, Modelling Simul. Mater. Sci. Eng. 33 (2025) 025007 (16 pages).
https://doi.org/10.1088/1361-651X/ada784

116. E. Habbershaw, R.S. Glasby, J.R. Haack, C.D. Hauck, and **S.M. Wise**, *Asymptotic Relaxation of Moment Equations for a Multi-species, Homogeneous BGK Model*, SIAP (SIAM J. Appl. Math.) 85 (2025) 294–313.
https://doi.org/10.1137/23M1596314

117. T. Luong and **S.M. Wise**, *A Nonnegative Weak Solution to the Phase Field Crystal Model with Degenerate Mobility*, Discrete Cont. Dyn. Syst. Series B 30 (2025) 3288–3309.
https://doi.org/10.3934/dcdsb.2025023

118. A. Christlieb, K. Promislow, Z. Tan, S. Wang, B. Wetton, **S.M. Wise**, *Benchmark Computation of Morphological Complexity in the Functionalized Cahn-Hilliard Gradient Flow*, Commun. Comput. Phys. 37 (2025) 877–920.
https://doi.org/10.4208/cicp.OA-2023-0190

119. E. Habbershaw, C.D. Hauck, and **S.M. Wise**, *Implicit Update of the Moment Equations for a Multi-Species, Homogeneous BGK Model*, SINUM (SIAM J. Numer. Anal.) 63 (2025) 881–907.
https://doi.org/10.1137/24M165421X

120. A.E. Diegel, C. Wang, and **S.M. Wise**, *Convergence Analysis of a Preconditioned Steepest Descent Solver for the Cahn-Hilliard Equation with Logarithmic Potential*, Int. J. Numer. Anal. Model. 22 (2025) 483–509.
https://doi.org/10.4208/ijnam2025-1021

121. T. Luong, T. Mengesha, **S.M. Wise**, and M.H. Wong, *A Diffuse Domain Approximation with Transmission-Type Boundary Conditions I: Asymptotic Analysis and Numerics*, Int. J. Numer. Anal. Model. 22 (2025) 694–727.
https://doi.org/10.4208/ijnam2025-1030

122. T. Luong, T. Mengesha, **S.M. Wise**, and M.H. Wong, *A Diffuse Domain Approximation with Transmission-Type Boundary Conditions II: Gamma Convergence*, Int. J. Numer. Anal. Model. 22 (2025) 728–744.
https://doi.org/10.4208/ijnam2025-1031

123. Y. Guo, C. Wang, **S.M. Wise**, and Z. Zhang, *A Uniquely Solvable and Positivity-Preserving Finite Difference Scheme for the Flory-Huggins-Cahn-Hilliard Equation with Dynamical Boundary Condition*, J. Comput. Appl. Math. 472 (2026) 116789 (21 pages).
https://doi.org/10.1016/j.cam.2025.116789

124. S. Tang, **S.M. Wise**, J.S. Lowengrub, and Z. Guo, *Membrane Heterogeneity Driven Dynamics of Multicomponent Vesicles in Shear Flow*, J. Fluid. Mech. 1030 (2026) A47 (36 pages).
https://doi.org/10.1017/jfm.2026.11276

125. J. Meng, S. Tang, **S.M. Wise**, and Z. Guo, *A GPU-Accelerated Matrix-Free FAS Multigrid Solver for Navier-Stokes Equations with Memory-Efficient Implementations*, Comput. Phys. Commun. 327 (2026) 110274.
https://doi.org/10.1016/j.cpc.2026.110274

126. Z. Zhang, S. Li, J.S. Lowengrub, and **S.M. Wise**, *An Efficient Constant-Coefficient MSAV Scheme for Computing Vesicle Growth and Shrinkage*, Commun. Comput. Phys. (online first, 2026).
https://doi.org/10.4208/cicp.OA-2026-0008

127. J.H. Park, A.J. Salgado, and **S.M. Wise**, *A Perturbed Preconditioned Gradient Descent Method for the Unconstrained Minimization of Composite Objectives*, J. Sci. Comput. (in review).
https://doi.org/10.48550/arXiv.2512.19532

128. T. Luong, T. Mengesha, K. Stinson, **S.M. Wise**, and M.H. Wong, *The α-Limit Problem: Convergence of a Linear Degenerate Interface Transmission Problem*, SIAM J. Appl. Math. (in review).
https://doi.org/10.48550/arXiv.2609.01237

Invited Research Presentations

1. Fourth SIAM Conference on Mathematical Aspects of Materials Science, Los Angeles, CA (May 2004). Research talk entitled *Phase Field Simulation of Quantum Dot Formation in a Thin Film*.
2. University of Virginia MRSEC/Lund University meeting, Lund, Sweden (July 2004). Research talk entitled *Phase Field Simulation of Quantum Dot Formation in a Thin Film*.
3. TMS Annual Meeting, The Armen G. Khachaturyan Symposium on Phase Transformation and Microstructural Evolution in Crystalline Solids, San Francisco, CA (February 2005). Research talk entitled *3D Phase Field Simulation of Q-Dot Formation (in a Strained Heteroepitaxial Film)*.
4. University of Buffalo, Mathematics Department, Buffalo, NY (January 2006). Colloquium talk entitled *Diffuse-Interface/Adaptive-Multigrid Calculations of Crystal Growth with Strong Interfacial Anisotropy*.
5. Research Center Caesar, Crystal Growth Group, Bonn, Germany (August 2006). Seminar talk entitled *Diffuse-Interface/Adaptive-Multigrid Calculations of Crystal Growth with Strong Interfacial Anisotropy*.
6. SIAM Conference on Nonlinear Waves and Coherent Structures (NW06), Seattle WA (September 2006). Research talk entitled *Continuum Models of Tumor Growth and Angiogenesis*.
7. University of Akron, Department of Theoretical and Applied Mathematics, Akron, OH (December 2006). Colloquium talk entitled *Modeling Solid Tumor Growth: Role of the Microstructure*.
8. Purdue University, Mathematics Department, West Lafayette, IN (January 2007). Colloquium talk entitled *Modeling Solid Tumor Growth: Role of the Microstructure*.
9. University of Louisville, Mathematics Department, Louisville, KY (January 2007). Colloquium talk entitled *Modeling Solid Tumor Growth: Role of the Microstructure*.
10. Ohio State University, Mathematics Department and Mathematical Biosciences Institute, Columbus, OH (February 2007). Colloquium talk entitled *Modeling Solid Tumor Growth: Role of the Microstructure*.
11. University of Tennessee, Mathematics Department, Knoxville, TN (February 2007). Colloquium talk entitled *Modeling Solid Tumor Growth: Role of the Microstructure*.
12. British Applied Mathematics Colloquium, Bristol, UK (April 2007). Research talk entitled *Computational Results for a Class of Regularized, Strongly Anisotropic Cahn-Hilliard Equations*.
13. Florida State University, Mathematics Department, Tallahassee, FL (September 2007). Colloquium talk entitled *The Diffuse Interface Method: Applications in Tumor Growth and Phase Transformations*.
14. SIAM Conference on Mathematical Aspects of Materials Science, Philadelphia, PA (May 2008). Research talk entitled *Large-Scale Coarsening Dynamics of Four-Fold Anisotropic Thin Films*.
15. Pennsylvania State University, Mathematics Department, University Park, PA (March 2009). Seminar talk entitled *Energy Stable and Convergent Schemes for the Phase Field Crystal (PFC) and Modified Phase Field Crystal (MPFC) Equations*.
16. 33rd SIAM Southeastern-Atlantic Section Conference, University of South Carolina, Columbia, SC (April 2009). Research talk entitled *Energy Stable and Convergent Schemes for the Phase Field Crystal (PFC) and Modified Phase Field Crystal (MPFC) Equations*.
17. 24th Annual Shanks Lecture and Conference, Vanderbilt University, Nashville, TN (May 2009). Plenary talk entitled *Multiscale Modeling of Solid Tumor Growth and Angiogenesis: The Effect of the Microenvironment*.

18. SIAM Conference on Mathematics for Industry: Challenges and Frontiers, San Francisco, CA (October 2009). Research talk entitled *Energy Stable and Convergent Schemes for the Phase Field Crystal (PFC) and Modified Phase Field Crystal (MPFC) Equations*.
19. University of Massachusetts, Mathematics Department, North Dartmouth, MA (November 2009). Seminar talk entitled *Energy Stable and Convergent Schemes for Bistable Gradient Equations*.
20. University of Pittsburgh, Mathematics Department, Pittsburgh, PA (January 2010). Colloquium talk entitled *Convex Splitting Schemes for Conserved Bistable Gradient Equations: Applications in Grain Boundary Motion, Solidification, and Tumor Growth*.
21. Purdue University, Mathematics Department, West Lafayette, IN (March 2010). Seminar talk entitled *Convex Splitting Schemes for Conserved Bistable Gradient Equations: Applications in Grain Boundary Motion, Solidification, and Tumor Growth*.
22. AMS Southeastern Section Meeting, University of Kentucky, Lexington, KY (March 2010). Research talk entitled *Energy Stable and Convergent Schemes for the Phase Field Crystal (PFC) and Modified Phase Field Crystal (MPFC) Equations*.
23. SIAM Annual Meeting, Pittsburgh, PA (July 2010). Research talk entitled *An Unconditionally Stable Second-Order Scheme for the Slope Selection Equation and Related Epitaxial Thin Film Models*.
24. SIAM Conference on the Life Sciences, Pittsburgh, PA (July 2010). Research talk entitled *Stable Finite Difference, Adaptive Nonlinear Multigrid Simulation of a Diffuse-Interface Model for Tumor Growth*.
25. University of South Carolina, Columbia SC (October 2010). Seminar talk entitled *Numerical and PDE Analyses of a Diffuse-Interface Model for Tumor Growth*.
26. Central Michigan University, Mount Pleasant, MI (March 2011). Seminar talk entitled *Numerical and PDE Analyses of a Diffuse-Interface Model for Tumor Growth*.
27. 35th SIAM Southeastern-Atlantic Section Conference, University of North Carolina, Charlotte, NC (March 2011). Research talk entitled *Second-Order Convex-Splitting Schemes*.
28. SIAM Conference on Applications of Dynamical Systems, Snowbird, UT (May 2010). (Declined invitation.)
29. 2011 International Conference on Applied Mathematics and Interdisciplinary Research, Nankai University, Tianjin, China and Beijing, China (June 2011). (Declined invitation.)
30. ICIAM, Vancouver, BC, Canada (July 2011). Research talk entitled *Second-Order Convex-Splitting Schemes*. (Declined invitation because of a scheduling conflict.)
31. University of South Carolina, Columbia, SC (January 2012). Research talk entitled *PDE/Numerical Analyses and Computations of Some Diffuse-Interface Models of Viscous Two-Phase Flows*.
32. Florida State University, Tallahassee, FL (February 2012). Research talk entitled *PDE/Numerical Analyses and Computations of some Diffuse Interface Models of Viscous Two-Phase Flows*.
33. Illinois Institute of Technology, Chicago, IL (March 2012). Research talk entitled *PDE/Numerical Analyses and Computations of Some Diffuse-Interface Models of Viscous Two-Phase Flows*.
34. ORNL/UTK Numerical Day 2012, Oak Ridge National Lab, Oak Ridge, TN (April 2012). Research talk entitled *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for the Cahn-Hilliard Equation*.
35. University of California, Irvine, CA (May 2012). Research talk entitled *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for a Modified Cahn-Hilliard Equation*.
36. Workshop on *Heterostructured Nanocrystalline Materials*, ICERM, Brown University, Providence, RI (May 2012). (Declined invitation because of a scheduling conflict.)
37. AIMS 9th International Conference on Dynamical Systems and Differential Equations, Orlando, FL (July 2012). Research talk entitled *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for the Cahn-Hilliard Equation*.
38. George Mason University, Fairfax, VA (September 2012). Research talk entitled *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for the Cahn-Hilliard Equation*.
39. Virginia Tech, Blacksburg, VA (September 2012). Research talk entitled *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for the Cahn-Hilliard Equation*.
40. *Workshop II: Atomistic and Mesoscale Modeling of Materials Defects*, IPAM's Program on *Materials Defects: Mathematics, Computation, and Engineering*, University of California, Los Angeles, CA (October 2012). Research talk entitled *Some Numerical Methods for PFC and DDFT-Like Equations*.

41. *International Workshop on Acoustic Activation of Surface Processes*, Breckenridge, CO (January 2013). Sponsored by DARPA. (Unable to attend due to a late scheduling conflict.)
42. SIAM Conference on Computational Science and Engineering, Boston, MA (February 2013). Research talk entitled *A Mixed Discontinuous Galerkin, Convex Splitting Scheme for the Cahn-Hilliard Equation*.
43. 37th SIAM Southeastern-Atlantic Section Conference, University of Tennessee, Knoxville, TN (March 2013). Research talk entitled *Second-Order Convex-Splitting Linear Iteration Scheme for Thin Film Models*.
44. University of Virginia, Charlottesville, VA (June 2013). Research talk entitled *The Phase Field Crystal Model: A Continuum Framework for Studying Phase Transformations on Atomic Length and Diffusive Time Scales*.
45. Oak Ridge National Lab, Computational Mathematics Seminar, Oak Ridge, TN (March 2014). Research talk entitled *Convergence of a Mixed Finite Element Method for a Cahn-Hilliard-Stokes System*.
46. University of North Carolina, Greensboro, NC (March 2014). Research talk entitled *Convergence of a Mixed Finite Element Method for a Cahn-Hilliard-Stokes System*.
47. Soochow University, Suzhou, China (June 2014). Research talk entitled *Convergence of a Mixed Finite Element Method for a Cahn-Hilliard-Stokes System*.
48. Fudan University, Shanghai, China (June 2014). Research talk entitled *Convergence of a Mixed Finite Element Method for Some Cahn-Hilliard-Flow Models*.
49. Shanghai Jiao Tong University, Shanghai, China (June 2014). Research talk entitled *Convergence of a Mixed Finite Element Method for Some Cahn-Hilliard-Flow Models*.
50. SIAM Annual Meeting, Chicago, IL (July 2014). Research talk entitled *Optimal Energy Norm Error Estimates for a Mixed FEM for a Cahn-Hilliard-Stokes System*.
51. SIAM Annual Meeting, Chicago, IL (July 2014). Research talk entitled *Stable and Convergent Numerical Schemes for Phase Field Crystal Models*.
52. Virginia Tech, Blacksburg, VA (October 2014). Research talk entitled *The Phase Field Crystal Model: A Continuum Framework for Studying Phase Transformations on Atomic Length and Diffusive Time Scales*.
53. Penn State, State College, PA (October 2014). Research talk entitled *The Phase Field Crystal Model: A Continuum Framework for Studying Phase Transformations on Atomic Length and Diffusive Time Scales*.
54. AMS Southeastern Section Meeting, Memphis, TN (October 2015). Research talk entitled *Convergence of a Second-order-in-time Finite Element Scheme for the Cahn-Hilliard Equation*.
55. SIAM SEAS 2016, Athens, GA (March 2016). Research talk entitled *Stability and Convergence for some Numerical Approximations of Cahn-Hilliard-Navier-Stokes Systems*.
56. Florida State University, Tallahassee, FL (March 2016). Research talk entitled *A Convergent, Energy Stable, and Efficient Hexagonal Finite Difference Scheme for the Phase Field Crystal Amplitude Expansion Model*.
57. Michigan State University, East Lansing, MI (April 2016). Research talk entitled *Higher-Order Convex Splitting Schemes for Bi-Stable Gradient Equations: Analysis and Practical Implementation*.
58. AIMS 11th International Conference on Dynamical Systems and Differential Equations, Orlando, FL (July 2016). Research talk entitled *A Convergent, Energy Stable, and Efficient Hexagonal Finite Difference Scheme for the Phase Field Crystal Amplitude Expansion Model*.
59. AIMS 11th International Conference on Dynamical Systems and Differential Equations, Orlando, FL (July 2016). Research talk entitled *A Multiphase Thin Film Model for Polymer Organic Photovoltaic (OPV) Device Preparation*.
60. Penn State, State College, PA (August 2016). Research talk entitled *A Multi-Component Model for Solvent Evaporation in a Polymer Thin Film*.
61. Virginia Tech, Blacksburg, VA (November 2016). Research talk entitled *Preconditioned Steepest Descent Methods for some Nonlinear Elliptic Equations Involving p-Laplacian Terms*.
62. University of North Carolina, Chapel Hill, NC (March 2017). Research talk entitled *Diffuse Interface Models of Density Mismatched, Multi-Phase Flows and Their Approximation Using Adaptive Multigrid Methods*.
63. University of California, Irvine, CA (April 2017). Research talk entitled *Preconditioned Steepest Descent Methods for some Nonlinear Elliptic Equations Involving p-Laplacian Terms*.

64. Oak Ridge National Laboratory, Computational and Applied Mathematics Seminar, Oak Ridge, TN (June 2017). Research talk entitled *Diffuse Interface Models of Density Mismatched, Multi-Phase Flows and Their Approximation Using Adaptive Multigrid Methods*.
65. Technische Universität Dresden, The Dresden Fellow Lecture, Mathematics Department, Dresden, Germany (November 2017). Research talk entitled *Diffuse Interface Models of Density Mismatched, Multi-Phase Complex Fluids, Soft Matter, and Beer Bubbles*.
66. Tufts University, Medford, MA (March 2018). Research talk entitled *Preconditioned Gradient Descent Methods for some Nonlinear Elliptic Equations Involving p-Laplacian Terms*.
67. SIAM Conference on Nonlinear Waves, Anaheim, CA (June 2018). Research talk entitled *A Phase Field Crystal Model for Graphene and Some Benchmark Problems*.
68. Illinois Institute of Technology, Chicago, IL (October 2018). Research talk entitled *Convergence Analysis of a Generalized Full Approximation Storage Scheme for Convex Optimization Problems*.
69. University of Alabama, Tuscaloosa, AL (March 2019). Research talk entitled *Convergence Analysis of Fast Subspace Descent (FASD) Scheme for Convex Optimization Problems*.
70. University of California, Irvine, CA (April 2019). Research talk entitled *Convergence Analysis of Fast Subspace Descent (FASD) Scheme for Convex Optimization Problems*.
71. 2019 Midwest Numerical Analysis Day, Illinois Institute of Technology, Chicago, IL (April 2019). Research talk entitled *A New Thermodynamically Consistent Phase Field Crystal Model with Multiple Phases and Heat Transport*.
72. Missouri University of Science and Technology, Rolla, MO (May 2019). Research talk entitled *Convergence Analysis of Fast Subspace Descent (FASD) Scheme for Convex Optimization Problems*.
73. The Cahn-Hilliard Equation: Recent Advances and Applications, NSF/CBMS Conference (May 2019). Research talk entitled *Convergence Analysis of Fast Subspace Descent (FASD) Scheme for Convex Optimization Problems Especially for Cahn-Hilliard-Type Equations*.
74. CAIMS Annual Meeting, Whistler, BC, Canada (June 2019). Research talk entitled *Numerical Methods and Benchmark Computations for the Functionalized Cahn-Hilliard Equation*. (Unable to attend due to a late flight cancellation. Talk given by a co-author.)
75. The University of Pittsburgh, Pittsburgh, PA (September 2019). Research talk entitled *Convergence Analysis of Fast Subspace Descent (FASD) Scheme for Convex Optimization Problems*.
76. Illinois Institute of Technology, Chicago, IL (March 2023). Research talk entitled *A Doubly Degenerate Diffuse-Interface Model of Surface Diffusion*.
77. SIAM Northern States Section Meeting, Logan, UT (April 2023), Plenary talk entitled *Doubly Degenerate Cahn-Hilliard Models of Surface Diffusion*.
78. ABPDE5 (Asymptotic Behaviors of Systems of PDEs Arising in Physics and Biology, 5th Edition), Lille, France (June 2023), Plenary talk entitled *Doubly Degenerate Cahn-Hilliard Models of Surface Diffusion*.
79. The University of Pittsburgh, Pittsburgh, PA (September 2023). Research talk entitled *Doubly Degenerate Cahn-Hilliard Equation: Asymptotic Analysis and Gamma Convergence*.
80. ORNL/UTK Phase-Field Workshop, Oak Ridge National Laboratory, Oak Ridge, TN (November 2023). Research talk entitled *Non-Isothermal Phase-Field-Crystal Model with Lattice Expansion*.
81. The University of California, Irvine, CA (December 2023). Research talk entitled *Non-Isothermal Phase Field Crystal Model with Lattice Expansion*.
82. The Ohio State University, Columbus, OH (January 2024). Research talk entitled *Non-Isothermal Phase Field Crystal Model with Lattice Expansion*.
83. SIAM Conference on Mathematics in Materials Science, Pittsburgh, PA (May 2024). Research talk entitled *Non-Isothermal Phase-Field-Crystal Model with Lattice Expansion*.
84. International Workshop on Advancements in Mesoscale Materials Modeling: Cutting-Edge Formulations, Simulations, and Applications, Technische Universität Dresden, Germany (December 2024). Research talk entitled *Non-Isothermal Phase Field Crystal Model with Lattice Expansion*.
85. The University of Washington, Seattle, WA (May 2025). Applied Mathematics Seminar. Research talk entitled *Gamma-Convergence and Asymptotic Analysis for a Diffuse Domain Problem with Transmission Boundary Conditions*.

86. The University of Pittsburgh, Pittsburgh, PA (October 2025). Computational Mathematics Seminar. Research talk entitled *Gamma-Convergence and Asymptotic Analysis for a Diffuse Domain Problem with Transmission Boundary Conditions*.

General-Audience Presentations

1. Tennessee Governor's School, October 2010. Presentation: *How to Become a Mathematician, and Maybe What to Do After That*.
2. University of Tennessee Pregame Faculty Showcase, November 2013. Presentation: *Simulations for Solutions: Solving Problems Through Scientific Computing*.
3. University of Tennessee Governor's School for the Sciences and Engineering, June 2016. Presentation: *Solving Problems with Scientific Computing, or Computing with Floating-Point Numbers*.
4. University of Tennessee high-school teacher-training event, *Let's Get Physical! Teaching Mathematics through the Lens of Physics*, July 2016. Presentation: *Modeling Soft-Matter Materials*.

Recent Teaching

Math 537, Continuum Mechanics, Fall 2026.
Math 579, Non-Equilibrium Thermodynamics, Spring 2026.
Math 574, Finite Element Methods, Fall 2025.
Math 674, Advanced Numerical PDE II (Multigrid Methods), Spring 2025.
Math 673, Advanced Numerical PDE I (Multigrid Methods), Fall 2024.
Math 515, Perturbation Theory and Asymptotic Analysis, Spring 2024.
Math 572, Numerical Mathematics II, Spring 2024.
Math 571, Numerical Mathematics I, Fall 2023.

Service

All university, college, and departmental service was performed at the University of Tennessee, Knoxville, unless otherwise indicated.

University and College Service

Department Head Exploratory Task Force for Physics, College of Arts and Sciences, 2025-26.

College Curriculum Committee, College of Arts and Sciences, 2025-26, 2026-27.

Electrical Engineering and Computer Science Department External Review Committee, Office of the Provost, Spring 2019.

Connections and Global Challenges Curriculum Committee, College of Arts and Sciences, 2016-17 through 2018-19.

Departmental Leadership and Service

Graduate Student Advising and Mentoring Committee, chair, 2022-23 through 2025-26.

Graduate Committee, 2007-08 through 2011-12, 2015-16, 2016-17, 2018-19, 2020-21, and 2022-23 through 2026-27; chair in 2010-11, 2011-12, Spring 2019, and 2022-23 through 2024-25.

Task Force on Reduced Teaching Loads, chair, Spring 2025.

Advisory Committee, 2011-12, 2012-13, 2014-15, 2015-16, 2018-19 (elected member), and 2019-20.

Faculty Search Committees, 2008-09 (Joint Institute for Computational Sciences), 2011-12 (Partial Differential Equations), 2012-13 (Computational and Applied Mathematics), 2016-17 and 2022-23 (Mathematical Biology), and 2023-24 (Computational and Applied Mathematics and Actuarial Mathematics).

Tenure and Promotion Committees, 2016-17 (A. Salgado, tenure and promotion), 2017-18 (T. Mengesha, tenure and promotion), 2020-21 (A. Salgado, promotion), and 2021-22 (T. Mengesha, promotion; O. Prosper, tenure and promotion), 2021-22 (O. Prosper, tenure and promotion), 2026-27 (O. Prosper, promotion).

Colloquium Committee, 2007-08 through 2009-10; chair in 2008-09 and 2009-10.

Assistantship Committee, 2012-13 and 2017-18.

Bylaws Committee, 2013-14, 2026-27; chair in 2026-27.

Student Mentoring, Examinations, and Outreach

NIMBioS Mentor, REU mentor in Summers 2009 and 2010; postdoctoral mentor for Sharon Bewick, August 2009 – May 2011; GRA advisor for Tri Sri Noor Asih, August 2013 – May 2014.

M.S. Examination Committees, nine committees, 2010-17; primary advisor for Kai Kang (Spring 2012) and Stephen E. Galloway (Spring 2017).

Ph.D. Examination Committees, three to five committees per year, ongoing.

UTK High School Mathematics Contest, timer in 2007, 2008, 2010, 2011, and 2014; judge in 2009, 2011, 2012, 2013, and 2015.

Editorial and Reviewing Service

Associate Editor, International Journal of Numerical Analysis and Modeling, 2024 – present.

Associate Editor, Advances in Applied Mathematics and Mechanics (AAMM), 2018 – present.

Associate Editor, Numerical Mathematics: Theory, Methods and Applications (NMTMA), 2017 – present.

Associate Editor, Discrete and Continuous Dynamical Systems, Series S (DCDS-S), 2014 – present.

Advisory Editor and Communicator, Mathematical Methods in the Applied Sciences (MMAS), 2014 – 2019.

NSF Panel Reviewer, Computing and Communication Foundations, Division of Materials Research, and Division of Mathematical Sciences.

Journal Reviewer, reviewer for many journals, including Mathematics of Computation, SIAM Journal on Numerical Analysis, Journal of Computational Physics, Physical Review E, and Mathematical Biosciences and Engineering.

Conference and Workshop Leadership

Conference Co-organizer, 2025 Section Meeting of the Southeastern Atlantic Section of the Society for Industrial and Applied Mathematics (SIAM), with V. Maroulas (lead organizer) and several co-organizers, March 21–23, 2025, Knoxville, Tennessee.

Conference Co-organizer, 2019 Annual Meeting of the Southeastern Atlantic Section of SIAM, with C. Webster (lead organizer) and G. Zhang, September 20–22, 2019, Knoxville, Tennessee.

Conference Co-organizer, 2017 Barrett Lectures: Mathematical Foundations of Data Science, with V. Maroulas (lead organizer), C. Webster, and J. Rosinski, May 1–3, 2017, Knoxville, Tennessee.

Conference Co-organizer, Southeastern-Atlantic Regional Conference on Differential Equations (SEARCDE), with S. Lenhart (lead organizer) and T. Phan, September 21–22, 2013, Knoxville, Tennessee.

Workshop Organizer, NIMBioS Investigative Workshop “Solid Tumor Modeling," January 19–21, 2011.

Steering Committee Member, Southeastern-Atlantic Regional Conference on Differential Equations (SEARCDE), October 2010 – October 2014.

Minisymposium Organizer, eight sessions on phase-field modeling, crystal dynamics, multiphase flow, scientific computing, and structured membranes at conferences of SIAM; the International Congress on Industrial and Applied Mathematics (ICIAM); the American Institute of Mathematical Sciences (AIMS); and the American Mathematical Society (AMS), 2010–21.

Graduate Students

Former (Alphabetical by Last Name)

1. Andreas Aristotelous, Mathematics, UTK, Ph.D. 2011. Thesis: “Adaptive Discontinuous Galerkin Finite Element Methods for a Diffuse Interface Model of Biological Growth." (O. Karakashian, UTK, was the primary advisor.) First position: postdoctoral researcher at Duke University and SAMSI.
2. Charlotte Beckford, Mathematics, UTK, Ph.D. 2026. Thesis: “Thermodynamically Consistent Modeling of Salt-in-Polyelectrolyte Batteries." (R. Kumar was the technical point of contact at ORNL.) First position: US Food and Drug Administration.
3. Kylie Berry, Mathematics, UTK, Ph.D. 2019. Topic: “Energy-Based Blending Models in Peridynamics." (T. Mengesha, UTK, was the co-advisor, and P. Seleson was the technical point of contact at ORNL.) First position: Assistant Professor of Mathematics at Dalton State College, Dalton, GA.
4. Craig Collins, Mathematics, UTK, Ph.D. 2015. Thesis: “Domain Decomposition Methods for Discontinuous Galerkin Approximations of Elliptic Problems." (O. Karakashian, UTK, was the primary advisor.) First position: Assistant Professor of Mathematics at Murray State University, Murray, KY.
5. John Cummings, Mathematics, UTK, Ph.D. 2017. Topic: “Simulating Block Polymer-Solvent Mixtures for Organic Photovoltaic Devices." (R. Kumar was the technical point of contact at ORNL.) First position: Visiting Assistant Professor, Berry College, Rome, GA.
6. Amanda Diegel, Mathematics, UTK, Ph.D. May 2015. Thesis: “Numerical Analysis of Convex Splitting Schemes for Cahn-Hilliard and Coupled Cahn-Hilliard-Fluid-Flow Equations." First position: Postdoctoral Researcher, Department of Mathematics, Louisiana State University, Baton Rouge, LA.
7. Wenqiang Feng, Mathematics, UTK, Ph.D. 2017. Topic: “Linearly Preconditioned Nonlinear Solvers for Phase Field Equations Involving p-Laplacian Terms." First position: Data Scientist, DST Systems, Inc., Kansas City, MO.
8. Stephen Galloway, Mathematics, UTK, M.S. 2017. Project: “Numerical Simulation of Solidification with Strong Interface Anisotropy." Continued in the Ph.D. program at UTK.
9. Stephen Galloway, Mathematics, UTK, Ph.D. 2020. Thesis: “Graphene Formation Using the PFC Model with a Vapor Phase." First position: Assistant Professor, Department of Mathematics, St. Andrews University, Laurinburg, NC.
10. Zhen Guan, Mathematics, UTK, Ph.D. August 2012. Thesis: “Numerical Analysis of First- and Second-Order Unconditionally Energy-Stable Schemes for Nonlocal Cahn-Hilliard and Allen-Cahn Equations." First position: Visiting Assistant Professor, Department of Mathematics, University of California, Irvine, CA.
11. Evan Habbershaw, Mathematics, UTK, Ph.D. 2025. Project: “Numerical Methods for Multispecies BGK Models." (C. Hauck was the technical point of contact at ORNL.) First position: Visiting Assistant Professor, Department of Mathematics, Pennsylvania State University, State College, PA.
12. Ben Plumridge, Mathematics, UTK, Ph.D. 2025. Project: “Neural Network-Based Adaptive Filtering of the Spherical Harmonic Method in Radiation Transport." (Cory Hauck, ORNL, was the primary advisor.)
13. Kai Kang, Mathematics, UTK, M.S. May 2012. Project: “A Second-Order Accurate, Energy-Stable Scheme for the Cahn-Hilliard-Hele-Shaw System of Equations: Application to Tumor Growth." Subsequently completed a Ph.D. at UTK in 2017.
14. Jea-Hyun Park, Mathematics, UTK, Ph.D. 2021. Thesis: “Preconditioned Nesterov’s Accelerated Gradient Descent Method and Its Applications to Nonlinear PDE." (A. Salgado, UTK, was the co-advisor.) First position: Visiting Assistant Professor, Department of Mathematics, University of California, Santa Barbara, CA.

Current (Alphabetical by Last Name)

1. Alli Brophy, Mathematics, UTK, Ph.D. expected 2030. Project: “Numerical Methods for Kinetic Models." UTK advising responsibility: 100%.
2. Jordan Crowell, Mathematics, UTK, M.S. expected December 2026. Project: “Phase Field Crystal Models." UTK advising responsibility: 100%.
3. Sam Gruber, Mathematics, UTK, Ph.D. expected 2027. Project: “Phase Transformations in $SrCoO_x$ Materials for Neuromorphic Computing Architectures." (V. Starchenko and P. Ganesh are the technical points of contact at ORNL.) UTK advising responsibility: 100%.
4. Jack Haight, Mathematics, UTK, Ph.D. expected 2028. Project: “Numerical Methods for Kinetic Equations." (C. Hauck is the technical point of contact at ORNL.) UTK advising responsibility: 100%.
5. Suleman Shahid, Mathematics, UTK, Ph.D. expected 2028. Project: To be determined. (Abner Salgado, UTK, is the co-advisor.) UTK advising responsibility: 50%.

6. Calvin Wong, Mathematics, UTK, Ph.D. expected 2027. Project: “Numerical Methods and Asymptotic Analysis for Diffuse-Domain Problems." (C. Hauck is the technical point of contact at ORNL for a companion project.) UTK advising responsibility: 100%.

Postdoctoral Researchers

Former

Toai Luong, 2022–2025, Ph.D., University of Alabama. First position after UTK: Visiting Assistant Professor, Virginia Commonwealth University, Richmond, VA.

References

1. **Wenbin Chen,** Professor
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9. **Brian Wetton,** Professor
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